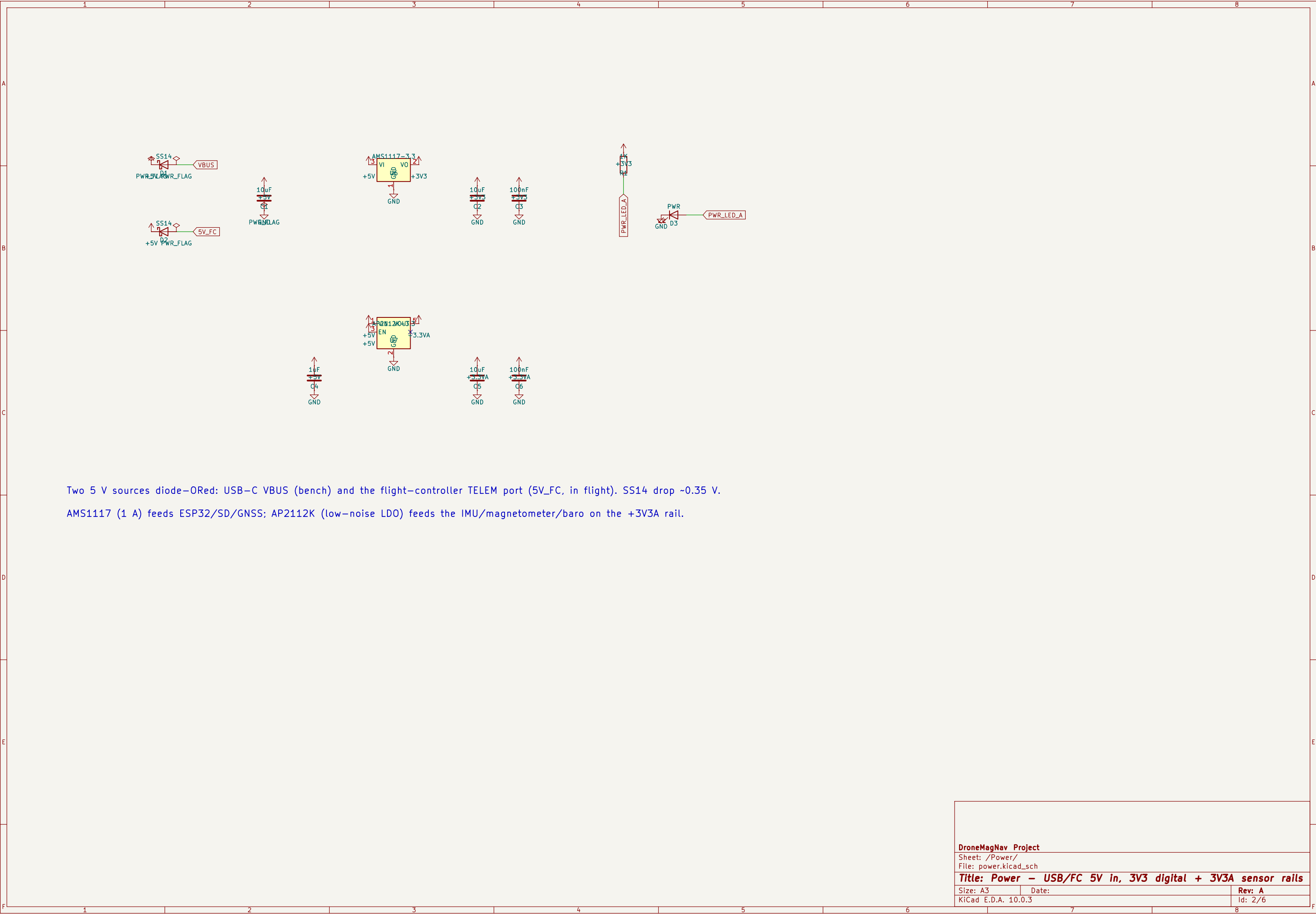
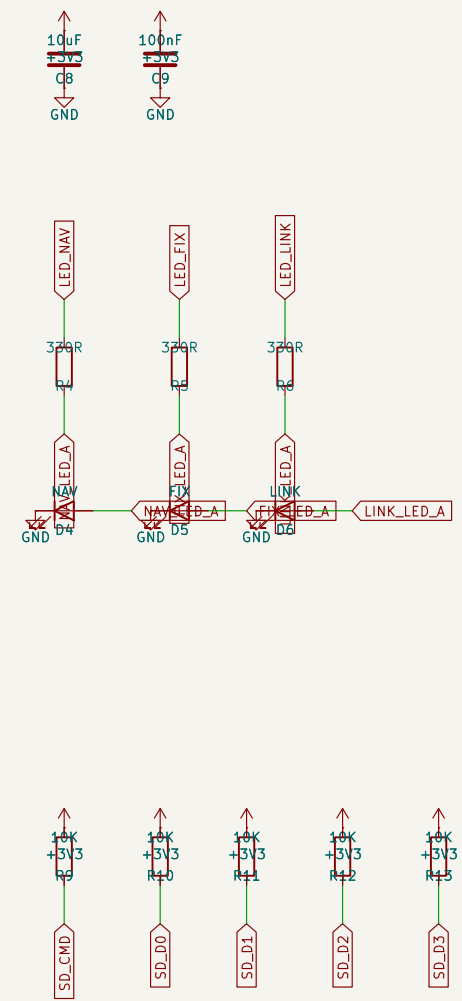


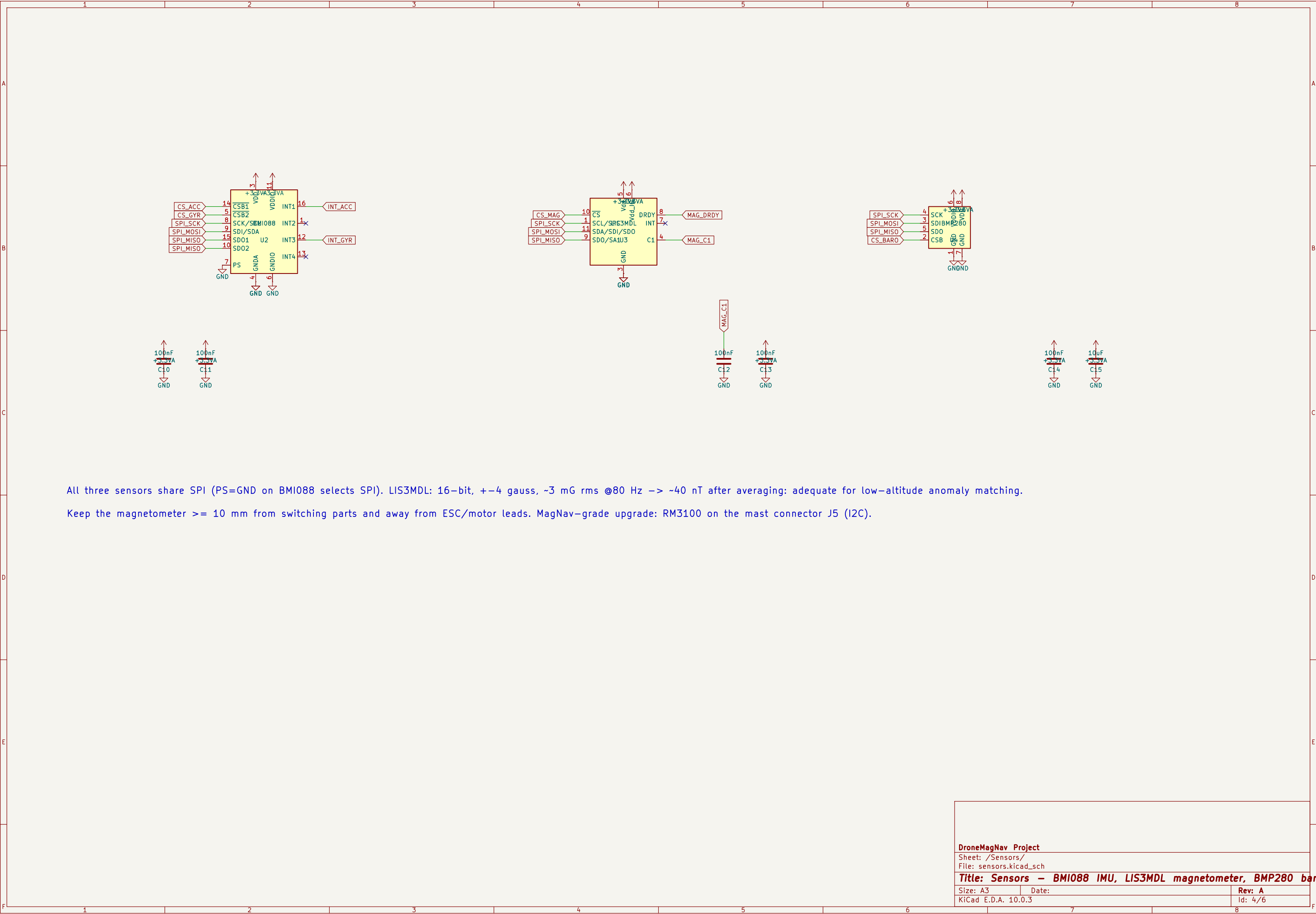




Two 5 V sources diode-ORed: USB-C VBUS (bench) and the flight-controller TELEM port (5V_FC, in flight). SS14 drop ~0.35 V.
AMS1117 (1 A) feeds ESP32/SD/GNSS; AP2112K (low-noise LDO) feeds the IMU/magnetometer/baro on the +3V3A rail.



SPI @10 MHz to IMU/mag/baro; SDMMC 4-bit to microSD; UART0 pads (GPIO43/44) GNSS; UART2 MAVLink to flight controller; I2C to external sensor mast.



All three sensors share SPI (PS=GND on BMI088 selects SPI). LIS3MDL: 16-bit, +-4 gauss, ~3 mG rms @80 Hz -> ~40 nT after averaging: adequate for low-altitude anomaly matching.

Keep the magnetometer >= 10 mm from switching parts and away from ESC/motor leads. MagNav-grade upgrade: RM3100 on the mast connector J5 (I2C).

